Etienne Lanzeray

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LinkedIn
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INTERESTS

Mathematics and Machine Learning.
Planning to pursue a PhD in ML surrogates for scientific simulations: current work in Computational Fluid Dynamic (3D Navier–Stokes inverse problems, wake flows, supersonic flows, and biomedical flows);
Methods: PINNs, classical & homogeneous Galerkin Neural Networks; with uncertainty quantification.
I'm also interested in 3D rendering (Overview article).

SKILLS

Programming: Python (Pandas, Scikit-learn, PyTorch, TensorFlow), C, Rust, SQL.
Parallel & GPU: Open MPI, CUDA.
Languages: French — native; English — TOEIC 970/990 (2025).

EDUCATION

Université de Lille, Centrale Lille & IMT Nord Europe, Lille, France
Master of Science, Data Science (in english & research-oriented) — 2024–2026

M2 (2025–2026): High Performance Computing; Bayesian Learning; Reinforcement Learning; Natural Language Processing; Computer Vision; Fairness & Privacy in ML; Reading Group; Research Project

M1 (2024–2025): Probability; Statistics; Algorithms & Complexity; Optimization & Numerical Analysis; Machine Learning (Hands-on, Landscape, Deep Learning); Models for ML; Signal Processing & Functional Data Analysis; Research Project.

Le Wagon, France
Data Science Training (data visualization, analysis, ML, DL, MLOps) — 2022

CentraleSupélec, France
Digital Tech Year Bootcamp (Agile/SCRUM, front-end dev) — 2018

Université Paris Saclay (Magistère Jacques Hadamard), Orsay, France
Master 1, Pure Mathematics — 2017–2018

Université Paris Saclay (Magistère Jacques Hadamard), Orsay, France
Licence 3, Mathematics (Pure & Applied) — 2016–2017

Lycée Gustave Eiffel, Bordeaux, France
Classes préparatoires PCSI → PSI* → PSI* — 2013–2016

RESEARCH PROJECTS

Applying homogeneous Neural Networks to fluid mechanics models via homogeneous Galerkin projection — 2025–2026
Advisors: Andrei Polyakov (Inria & CRIStAL, CNRS), Mihaly Petreczky (CRIStAL, CNRS)

Performance evaluation and generalization bounds of homogeneous Neural Networks — 2024–2025
(Overview article)
Advisors: Andrei Polyakov (Inria & CRIStAL, CNRS), Mihaly Petreczky (CRIStAL, CNRS)

The analytic large sieve and applications to number theory — 2017–2018
Advisor: Etienne Fouvry (Institut de Mathématiques d'Orsay) (Overview article)

Uniqueness of trigonometric series expansions — 2016–2017
Advisor: Julien Sabin (Institut de Mathématiques d'Orsay)

EXPERIENCE

Research Intern — Université de Lille
2025 — Lille, France
Continuation of my research project on *Performance evaluation and generalization bounds of homogeneous Neural Networks*, advised by Andrei Polyakov (Inria & CRIStAL, CNRS) and Mihaly Petreczky (CRIStAL, CNRS)

Tutor in Computer Science (L1) — Université de Lille
2024–2025 — Lille, France
Supported first-year students outside class hours to complete tutorials and labs.
Topics: programming fundamentals, algorithms, and basic data structures.

Data Science Intern — Azur Drones
2023–2024 — Mérignac, France
Designed and put in production a flight database (with TimescaleDB). Built analytics for behavior and predictive maintenance. Trained an LSTM autoencoder for anomaly detection on ~ 80 features with $> 90\%$ accuracy.

Volunteer ML Collaborator — Ebiose (Inria Startup Studio)
2023–2024 — Bordeaux, France
Prototyped a contrastive-learning pipeline to select and score AI agents on *GSM8k*.
Combined Transformer text encoders with graph neural networks. (repo)

Private Teacher — Acadomia
2019–2020 — France
Taught mathematics and physics–chemistry to secondary students.

Oral Examiner (CPGE) — Lycée Gustave Eiffel
2017 — Bordeaux, France
Prepared students for French engineering school entrance exams.
Conducted 30–45 minute oral boards in mathematics.

Research Intern — ICMCB (CNRS) & CEA Saclay
2016 — Bordeaux & Saclay, France
Modeled the growth of nodules on silica nanoparticles. Built mathematical models and worked on computational tools for simulations.